

Description	Owners/Teams	Impact	Effort (FTE)	Milestones (M) and Decision Points (D) (year: description)	Risks and External Dependencies	Prob. of Applicability /Success/Adoption (Low, Medium, High)
3.11 Submission Infrastructure: Scalability	CMS ownership: SI (Antonio Perez-Calero, Marco Mascheroni)	Accommodate and exploit new, substantial resources in the CMS pool	1.0	Run simultaneously and efficiently: '21 M: 0.5 M jobs (achieved) '22 M: 0.75 M jobs '23 M: 1.0 M jobs	Dependencies: HTCondor, SI, CCB Risks: delays in the HTCondor software development, inadequate SI infrastructure, potential loss of expertise with replacement of main SI operator	Success: High
3.5 Evolution of Workflow Management	Development team: CMS CMS ownership: DMWM (K. Lannon, D. Ciangottini)	Enable increasingly heterogeneous and diverse set of resources (through flexibility, automation)	1.5 1.0 2.0	'22 D: evolutionary or substantial intervention '23 M: cost/benefit analysis for workflow description, task decomposition, data scheduling '24 D: overall design '26-'28 M: development, testing	Dependencies: HTCondor, Rucio, FTS, Coordination with SI, PdmV, framework. Risks: too fast evolution of HPC and accelerator technology, not enough experienced developers,	Success: High
3.12 Allocation aware campaign and production planning	CMS ownership: DRP (D. Spiga,	Allow well-suited resource allocation	1	'22-'23: gather requirements and	Dependencies:	Success: High
and management	C. Wissing)	depending on physics and groups priorities		dependencies. Prototype, evaluate initial results '24: Full implementation (new service or additional functionality for existing ones). Commissioning and early production. '25: Further development (DevOps), operators training	PdmV tools, data pre-placement/staging, WM tools Risks: Expertise availability, long-term support for dedicated service	
3.16 Architectures and platforms	Development team: CMS CMS ownership: DRP (D. Spiga, C. Wissing)	Enable allocations in HPCs with non-x86-64 CPUs	0.2	M 2022: Finalize POWER9 as 1st non-x86 architecture for production M 2023: GPU workflows (assuming to be available for production in '23) in the distributed infrastructure M 2023: Support for other non-x86 architecture, e.g. ARM, IBM	Dependencies: HPC hardware technology, submission infrastructure (HTCondor), SI, DMWM Risks: Existing system assumes mono-architecture, complex to make it more generic	Success: High